The Gaussian Heat Kernel as an Initial-Value Solution

An End-to-End Meta-Math Walkthrough

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Abstract

The Gaussian simultaneously solves the PDE and has total mass one. Its shrinking tails transfer uniform continuity of the initial data into uniform convergence. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PDE-04 / Partial Differential Equations / Report v1

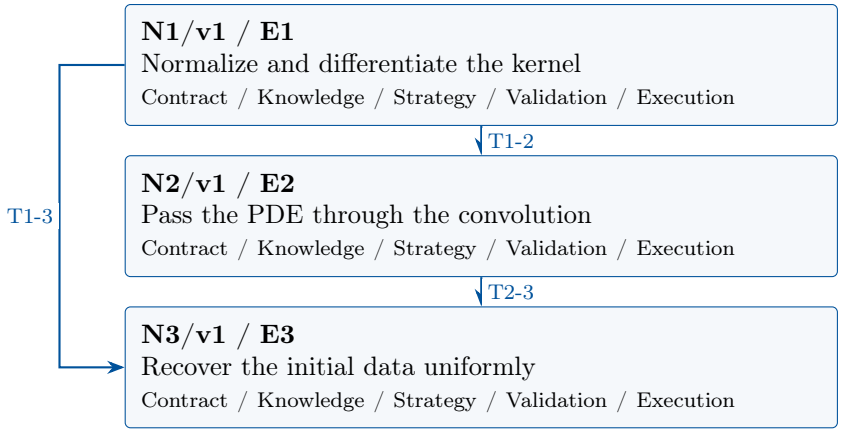
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For $\kappa > 0$ and bounded uniformly continuous $f : \mathbb{R}^d \to \mathbb{R}$, define $G_t(z) = (4\pi\kappa t)^{-d/2} \exp(-\|z\|^2/(4\kappa t))$ and $u(x,t) = \int_{\mathbb{R}^d} G_t(x-y) f(y)\, dy$ for $t > 0$. Prove this integral exists, u is smooth for $t > 0$, $u_t = \kappa \Delta u$, $\|u(\cdot,t)\|_\infty \le \|f\|_\infty$, and $u(\cdot,t) \to f$ uniformly as $t \downarrow 0$. This project constructs a solution and does not assert uniqueness among all unrestricted whole-space solutions.*

A. Foundations and permitted imports

Import the Gaussian integral $\int_{\mathbb{R}^d} e^{-\|z\|^2}\, dz = \pi^{d/2}$, Euclidean change of variables, dominated convergence, and differentiation under an integral when derivatives have a locally uniform integrable majorant. Polynomials times Gaussians are integrable; their tails tend to zero by integrability. Uniform continuity means one δ works for all spatial points.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $G_t > 0$, $\int G_t = 1$, and $\partial_t G_t = \kappa \Delta G_t$.

Receiving activity. N2.1

Compatibility. E1/v1 and bounded f. Check: Time neighborhoods must stay away from zero; derivative kernels become singular as $t \downarrow 0$.

T1-3: N1/E1/v1 to N3. $G_t > 0$, $\int G_t = 1$, and $\partial_t G_t = \kappa \Delta G_t$.

Receiving activity. N3.1

Compatibility. E1/v1 mass and bounded uniform continuity of f. Check: The same δ must work for every x; pointwise continuity does not provide a uniform conclusion.

T2-3: N2/E2/v1 to N3. u exists absolutely, is smooth for $t > 0$, satisfies the heat equation, and obeys the sup-norm bound.

Receiving activity. N3.1

Compatibility. E1/v1 mass and bounded uniform continuity of f. Check: The same δ must work for every x; pointwise continuity does not provide a uniform conclusion.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Normalize and differentiate the kernel

Object and role. Compute the mass by scaling and compare the time derivative with the spatial Laplacian. **Local obligation:** $G_t > 0$, $\int G_t = 1$, and $\partial_t G_t = \kappa \Delta G_t$.

Inputs. G_t for $t > 0$ and fixed $\kappa > 0$.

C — Contract

Publish E1/v1 only after establishing this obligation: $G_t > 0$, $\int G_t = 1$, and $\partial_t G_t = \kappa \Delta G_t$.

K — Knowledge

G_t for $t > 0$ and fixed $\kappa > 0$. Retain the stated assumptions and import versions.

S — Strategy

Compute the mass by scaling and compare the time derivative with the spatial Laplacian.

V — Validation

The coefficient $4\kappa t$ controls both the mass normalization and the PDE coefficient.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Set $z = \sqrt{4\kappa t}\, w$. The Gaussian integral gives $\int G_t(z)dz = \pi^{-d/2} \int e^{-\|w\|^2} dw = 1$. Positivity is immediate.

N1.2 Differentiation gives $\partial_t G_t = G_t(-d/(2t) + \|z\|^2/(4\kappa t^2))$ and $\Delta G_t = G_t(-d/(2\kappa t) + \|z\|^2/(4\kappa^2 t^2))$. These formulas prove the PDE identity. Higher derivatives are finite sums of polynomial factors times a Gaussian with powers of t in their coefficients.

Export E1/v1. $G_t > 0$, $\int G_t = 1$, and $\partial_t G_t = \kappa \Delta G_t$.

Acceptance gate. The coefficient $4\kappa t$ controls both the mass normalization and the PDE coefficient.

Targeted repair. If the kernel normalization or diffusivity disagrees, recompute both derivatives and the scaled mass together. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Pass the PDE through the convolution

Object and role. Bound convolution by kernel mass and justify differentiation on compact positive-time neighborhoods. **Local obligation:** u exists absolutely, is smooth for $t > 0$, satisfies the heat equation, and obeys the sup-norm bound.

Inputs. E1/v1 and bounded f. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: u exists absolutely, is smooth for $t > 0$, satisfies the heat equation, and obeys the sup-norm bound.

K — Knowledge

E1/v1 and bounded f. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Bound convolution by kernel mass and justify differentiation on compact positive-time neighborhoods.

V — Validation

Time neighborhoods must stay away from zero; derivative kernels become singular as $t \downarrow 0$.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Absolute integrability follows from $\int |G_t(x-y)f(y)|dy \le \|f\|_\infty \int G_t = \|f\|_\infty$, which also proves the sup-norm bound.

N2.2 Fix a compact set of x values and $0 < a \le t \le b < \infty$. Every fixed derivative of $G_t(x-y)$ is bounded there by a constant times a polynomial in $\|y\|$ times $e^{-c\|y\|^2}$, for some $c > 0$: bounded x and positive lower and upper time bounds permit absorbing shifts and powers into a slower Gaussian. Multiplying by bounded f gives an integrable majorant. Differentiation under the integral is therefore valid for every order locally. E1 then gives $u_t - \kappa\Delta u = \int(\partial_t G_t - \kappa\Delta G_t)f = 0$.

Export E2/v1. u exists absolutely, is smooth for $t > 0$, satisfies the heat equation, and obeys the sup-norm bound.

Acceptance gate. Time neighborhoods must stay away from zero; derivative kernels become singular as $t \downarrow 0$.

Targeted repair. If differentiation is justified by pointwise smoothness alone, supply the positive-time Gaussian majorant before passing the derivative through the integral. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Recover the initial data uniformly

Object and role. Split the convolution error into a uniformly small near part and a vanishing Gaussian tail. **Local obligation:** $\sup_x |u(x,t) - f(x)| \to 0$ as $t \downarrow 0$.

Inputs. E1/v1 mass and bounded uniform continuity of f. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: $\sup_x |u(x,t) - f(x)| \to 0$ as $t \downarrow 0$.

K — Knowledge

E1/v1 mass and bounded uniform continuity of f. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Split the convolution error into a uniformly small near part and a vanishing Gaussian tail.

V — Validation

The same δ must work for every x; pointwise continuity does not provide a uniform conclusion.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 Using mass one, write $u(x,t) - f(x) = \int G_t(z)(f(x-z) - f(x))dz$. Given $\varepsilon > 0$, choose $\delta > 0$ with $|f(x-z) - f(x)| < \varepsilon$ whenever $\|z\| < \delta$, uniformly in x. The integral over that region has absolute value at most ε.

N3.2 The remaining part has absolute value at most $2\|f\|_\infty \int_{\|z\| \geq \delta} G_t(z)dz$. Scaling expresses the last integral as $\pi^{-d/2} \int_{\|w\| \geq \delta/\sqrt{4\kappa t}} e^{-\|w\|^2} dw$, which tends to zero. The bound is independent of x, proving uniform convergence. Define $u(x,0) = f(x)$ to attach the initial data.

Export E3/v1. $\sup_x |u(x,t) - f(x)| \to 0$ as $t \downarrow 0$.

Acceptance gate. The same δ must work for every x; pointwise continuity does not provide a uniform conclusion.

Targeted repair. If a point-dependent continuity radius is used in a supremum estimate, invoke uniform continuity and repeat the split with one global radius. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Normalize and differentiate the kernel

1. **N1.1.** Set $z = \sqrt{4\kappa t}\, w$. The Gaussian integral gives $\int G_t(z)dz = \pi^{-d/2} \int e^{-\|w\|^2} dw = 1$. Positivity is immediate.

2. **N1.2.** Differentiation gives $\partial_t G_t = G_t(-d/(2t) + \|z\|^2/(4\kappa t^2))$ and $\Delta G_t = G_t(-d/(2\kappa t) + \|z\|^2/(4\kappa^2 t^2))$. These formulas prove the PDE identity. Higher derivatives are finite sums of polynomial factors times a Gaussian with powers of t in their coefficients.

B. N2 — Pass the PDE through the convolution

1. **N2.1.** Absolute integrability follows from $\int |G_t(x-y)f(y)|dy \le \|f\|_\infty \int G_t = \|f\|_\infty$, which also proves the sup-norm bound.

2. **N2.2.** Fix a compact set of x values and $0 < a \le t \le b < \infty$. Every fixed derivative of $G_t(x-y)$ is bounded there by a constant times a polynomial in $\|y\|$ times $e^{-c\|y\|^2}$, for some $c > 0$: bounded x and positive lower and upper time bounds permit absorbing shifts and powers into a slower Gaussian. Multiplying by bounded f gives an integrable majorant. Differentiation under the integral is therefore valid for every order locally. E1 then gives $u_t - \kappa\Delta u = \int (\partial_t G_t - \kappa\Delta G_t) f = 0$.

C. N3 — Recover the initial data uniformly

1. **N3.1.** Using mass one, write $u(x,t) - f(x) = \int G_t(z)(f(x-z) - f(x))dz$. Given $\varepsilon > 0$, choose $\delta > 0$ with $|f(x-z) - f(x)| < \varepsilon$ whenever $\|z\| < \delta$, uniformly in x. The integral over that region has absolute value at most ε.

2. **N3.2.** The remaining part has absolute value at most $2\|f\|_\infty \int_{\|z\|\ge\delta} G_t(z)dz$. Scaling expresses the last integral as $\pi^{-d/2} \int_{\|w\|\ge\delta/\sqrt{4\kappa t}} e^{-\|w\|^2} dw$, which tends to zero. The bound is independent of x, proving uniform convergence. Define $u(x,0) = f(x)$ to attach the initial data.

□

V. WORKED ORCHESTRATION

In one dimension with $\kappa = 1$ and $f(x) = e^{-x^2}$, completing the square inside the Gaussian convolution gives $u(x,t) = (1+4t)^{-1/2} \exp(-x^2/(1+4t))$. The Gaussian integral supplies the prefactor. This broadens and decreases in amplitude, tends to the original Gaussian as $t \downarrow 0$, and direct differentiation confirms $u_t = u_{xx}$.

A. Boundary and hypothesis audit

For discontinuous $f = 1_{(0,\infty)}$ in one dimension, symmetry gives $u(0,t) = 1/2$ for every $t > 0$, whereas $f(0) = 0$. The uniform initial-data conclusion is false. The convolution still exists and solves the PDE at positive time, illustrating which conclusion uses continuity.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N3 claim: mass one alone implies uniform recovery of every bounded f. **Validation witness:** the step function at the origin. **Diagnosis:** Strategy omitted the near-field continuity estimate. **Repair:** use bounded uniform continuity as explicitly assumed and the near/tail decomposition. Recheck only the initial-data export; the positive-time kernel and PDE calculations remain valid under boundedness.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Prove a uniqueness theorem in a specified bounded solution class, or extend convergence to L^p initial data using approximation.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Leon Simon, Lectures on PDE, for elliptic, weak, and parabolic methods. Classical calculus and the explicitly stated Hilbert-space imports are used below; the displayed PDE arguments are developed in full.